

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using fork() system call.
- b) Read the user input as shell command.
- c) Use a system call exec() to execute the shell command.
- d) Display the PIDs of parent and child processes.

```
UW PICO 5.09 File: task.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    // Read command from user
    printf("Enter a linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove newline character
    command[strcspn(command, "\n")] = '\0';

    // Create child process
    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    // Child process
    else if (pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        // Execute the command
        execlp(command, command, NULL);

        // Executes only if exec fails
        perror("Execution failed");
        exit(1);
    }

    // Parent process
    else {
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        // Wait for child process
        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

Get Help Exit WriteOut Justify Read File Where is Prev Pg Next Pg Cut Text UnCut Text Cur Pos To Spell

1)Then press ctrl+o enter ctrl+x2)then gcc filename.c -o filename

```
Last login: Fri Aug 7 14:25:49 on ttys000
sreenikhilreddy@Srees-MacBook-Pro ~ % nano.c
zsh: command not found: nano.c
sreenikhilreddy@Srees-MacBook-Pro ~ % nano task.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc task.c -o task
sreenikhilreddy@Srees-MacBook-Pro ~ % ./task
Enter a Linux command: ls

Parent Process
Parent PID : 1082
Child PID : 1087

Child Process
Child PID : 1087
Parent PID : 1082
1.c.save      3      a.out      dir2      Library      Parallels      python
12            3.c      Applications dir3      Movies      Pictures      task
12.c          4        Applications (Parallels) Documents    Music      PROJECT DSA 2    task.c
2             4.c      Desktop    Downloads  myfolder     projects      terminal
2.c           4.c.save dirl       helloworld NewFolderName Public

Child process completed.
sreenikhilreddy@Srees-MacBook-Pro ~ % nano task.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc ./task
zsh: no such file or directory: gcc./task
sreenikhilreddy@Srees-MacBook-Pro ~ % ./task
Enter a Linux command: pw

Parent Process
Parent PID : 1098
Child PID : 1099

Child Process
Child PID : 1099
Parent PID : 1098
Execution failed: No such file or directory

Child process completed.
sreenikhilreddy@Srees-MacBook-Pro ~ % ./task
Enter a Linux command: uname

Parent Process
Parent PID : 1103
Child PID : 1104

Child Process
Child PID : 1104
Parent PID : 1103
Darwin

Child process completed.
sreenikhilreddy@Srees-MacBook-Pro ~ % █
```

```
Last login: Fri Aug 7 14:25:49 on ttys000
sreenikhilreddy@Srees-MacBook-Pro ~ % nano.c
zsh: command not found: nano.c
sreenikhilreddy@Srees-MacBook-Pro ~ % nano task.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc task.c -o task
sreenikhilreddy@Srees-MacBook-Pro ~ % ./task
Enter a Linux command: ls

Parent Process
Parent PID : 1082
Child PID : 1087

Child Process
Parent PID : 1082
1.c.save      3      a.out      dir2      Library      Parallels      python
12            3.c      Applications dir3      Movies      Pictures      task
12.c          4        Applications (Parallels) Documents    Music      PROJECT DSA 2  task.c
2             4.c      Desktop    Downloads  myfolder     projects      terminal
2.c           4.c.save dirl       helloworld NewFolderName Public

Child process completed.
sreenikhilreddy@Srees-MacBook-Pro ~ % nano task.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc ./task
zsh: no such file or directory: gcc./task
sreenikhilreddy@Srees-MacBook-Pro ~ % ./task
Enter a Linux command: pw

Parent Process
Parent PID : 1098
Child PID : 1099

Child Process
Child PID : 1099
Parent PID : 1098
Execution failed: No such file or directory

Child process completed.
sreenikhilreddy@Srees-MacBook-Pro ~ % █
```

Child process completed.

Task 2

Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

An operating system is system software that controls all the hardware and software resources of a computer. Users do not communicate directly with the hardware. Instead, the operating system acts as a bridge between the user, applications, and the hardware

Linux Commands Used

Report: Investigating Hardware Resources and Operating System Services Using Linux Commands

Title

Investigation of Hardware Resources and Operating System Services Using Linux Terminal Commands

1. Aim

To study how the Linux operating system interacts with hardware resources using terminal commands and understand how the operating system abstracts CPU, memory, storage, and I/O devices.

2. Objective

- To use Linux commands such as `uname`, `lscpu`, `ps`, and `top`.
- To examine CPU and system information.
- To observe running processes.
- To understand how the operating system manages hardware resources.
- To explain how the OS provides an abstraction layer between users and hardware.

3. Introduction

An **Operating System (OS)** is system software that manages all hardware and software resources of a computer. Users and application programs do not directly communicate with hardware. Instead, the operating system acts as a bridge between the user, applications, and hardware devices.

The OS performs several important functions:

- Process Management

- Memory Management
- File Management
- Device Management
- Security
- User Interface

Linux provides several terminal commands that help users inspect hardware resources and operating system services.

1. uname

The `uname` command is used to display information about the Linux operating system. It provides details such as the operating system name, kernel version, machine architecture, and hostname. This command helps us identify the system on which Linux is running.

```
shashankreddy@Shashanks-MacBook-Air ~ % uname
Darwin
shashankreddy@Shashanks-MacBook-Air ~ %
```

2. lscpu

The `lscpu` command displays detailed information about the processor. It shows the CPU model, number of cores, number of threads, CPU architecture, and processor speed. This information helps us understand the hardware capabilities of the computer.

```
shashankreddy@Shashanks-MacBook-Air ~ % ./task2
Enter a Linux command: lscpu

Parent Process
Parent PID: 28740
Child PID: 28741

Child Process
Child PID: 28741
Parent PID: 28740
Invalid command!
Child process completed.
shashankreddy@Shashanks-MacBook-Air ~ %
```

3. ps

The `ps` command lists the processes that are currently running in the system. It displays the Process ID (PID), Parent Process ID (PPID), user name, and the command associated with each process. This command is useful for monitoring and managing processes.

```
shashankreddy@Shashanks-MacBook-Air ~ % ps
  PID TTY          TIME CMD
 28674 ttys000    0:00.05 -zsh
shashankreddy@Shashanks-MacBook-Air ~ %
```

4. top

The top command provides a real-time view of the system's performance. It continuously displays CPU usage, memory usage, running processes, and system load. It is commonly used to identify programs that consume a large amount of system resources.

PID	COMMAND	%CPU	TIME	#TH	#WQ	#PORT	MEM	PURG	CMPRS	PGRP	PPID	STATE	BOOSTS
591	WindowServer	41.2	04:24:16	23	5	4171	616M-	55M	87M	591	1	sleeping	*0[1]
0	kernel_task	9.1	02:18:20	617/10	0	0	39M	0B	0B	0	0	running	0[0]
28749	top	6.6	00:01.28	1/1	0	29	9872K	0B	0B	28749	28674	running	*0[1]
28750	screencaptur	5.3	00:00.30	3	2	78	6736K	16K	0B	1270	1270	sleeping	*0[665+]
28751	screencaptur	1.3	00:00.21	5	3	189	17M	0B	0B	28751	1	sleeping	*0[165+]
689	com.apple.Dr	1.2	29:05.65	8	6	1392	31M	0B	3232K	689	1	sleeping	0[1]
28239	Google Chrom	0.9	01:34.18	18	3	325-	474M-	51M	22M	28233	28233	sleeping	*1[30872]
652	mDNSResponde	0.8	19:20.13	3	1	121	9697K	0B	1136K	652	1	sleeping	*0[1]
28233	Google Chrom	0.6	01:17.39	44	3	800	175M	0B	35M	28233	1	sleeping	*17530[175]
28672	Terminal	0.5	00:03.85	10	5	320	196M+	42M	0B	28672	1	sleeping	*0[99+]
1101	ControlCentr	0.4	05:28.30	8	3	756	71M	384K	31M	1101	1	sleeping	*0[27699]
28581	netbiosd	0.4	00:01.20	6	5	30	2608K	0B	272K	28581	1	sleeping	*0[1]
28616	Google Chrom	0.3	00:04.83	29	1	702	172M+	0B	8256K	28233	28233	sleeping	*3-[1373+]
28631	WhatsApp	0.2	00:22.43	30	7	861	202M	19M	15M	28631	1	sleeping	*57[3]
578	corebrightne	0.1	03:03.21	3	2	170	5200K	0B	1504K	578	1	sleeping	*0[1]
24617	mysqld	0.1	00:29.51	38	0	61	494M	0B	476M	24514	24514	sleeping	*0[1]
28663	Google Chrom	0.1	00:07.77	23	1	243	245M	16K	0B	28233	28233	sleeping	*15[806]
922	distnoted	0.1	00:47.96	3	2	625	4176K	0B	816K	922	1	sleeping	*0[1]
565	distnoted	0.1	00:28.20	3	2	255	2432K	0B	768K	565	1	sleeping	*0[1]
524	powerd	0.1	01:50.66	5	4	167	5648K	0B	688K	524	1	sleeping	*0[1]
510	logd	0.0	05:50.13	4	3	2704	14M	0B	14M	510	1	sleeping	*0[1]
671	audioclocksy	0.0	01:17.52	3	2	51	2816K	0B	1168K	671	1	sleeping	*0[1]
575	bluetoothd	0.0	02:13.22	6	5	384	9825K	0B	3536K	575	1	sleeping	*0[1]
649	airportd	0.0	20:51.73	10	8	395	17M	0B	1936K	649	1	sleeping	*1[1]
580	com.apple.cm	0.0	00:04.69	4	3	188	15M	0B	2288K	580	1	sleeping	*0[3563+]
924	cfprebsd	0.0	01:42.23	3	2	866	4464K	3072K	1296K	924	1	sleeping	*33[80009]
28240	Google Chrom	0.0	00:09.47	19	1	149	42M	0B	16M	28233	28233	sleeping	*147-[405]
25734	PerfPowerSer	0.0	01:08.43	5	2	550-	13M-	416K	1776K	25734	1	sleeping	0[1983]
1190	duetexpertd	0.0	16:30.61	2	1	422	19M	1200K	3824K	1190	1	sleeping	0[10464]
545	coreduetd	0.0	00:30.05	9	8	142-	8497K	6496K	1824K	545	1	sleeping	*16[8979]
1001	suggestd	0.0	01:26.75	7	6	556	17M	192K	5152K	1001	1	sleeping	0[2471]
945	universalacc	0.0	00:10.98	8	3	255	8929K	0B	4544K	945	1	sleeping	*0[22654+]
1271	Finder	0.0	04:41.44	9	5	1416	361M	3184K	220M	1271	1	sleeping	*0[9758]
1038	fileprovider	0.0	35:24.21	3	2	175	42M	624K	25M	1038	1	sleeping	*3582[52]
832	mds_stores	0.0	10:40.85	5	3	139	28M	144K	5392K	832	1	sleeping	*0[1]
625	symptomsd	0.0	02:05.76	4	3	195	7680K	128K	1536K	625	1	sleeping	0[34483]
976	lockoutagent	0.0	00:18.84	3	2	59	3104K	0B	928K	976	1	sleeping	*0[1]
728	distnoted	0.0	00:03.51	3	2	31	1744K	0B	912K	728	1	sleeping	*0[1]
626	distnoted	0.0	00:04.21	3	2	37	1840K	0B	912K	626	1	sleeping	*0[1]
977	sharingd	0.0	01:37.74	4	1	387	23M	0B	10M	977	1	sleeping	*0[1]
642	distnoted	0.0	00:03.76	3	2	31	1728K	0B	896K	642	1	sleeping	*0[1]